

1. Equilibrium / 1 bath / finite temp. Ngeturbre.

→ model: $n := n_a + n_b$, $m := n_a - n_b$; $n=1$ fixed, $t_a = -t_b$

$$\begin{pmatrix} \epsilon_a(k) + \frac{V}{2}n - \frac{V}{2}m & -\Delta^* \\ -\Delta & \epsilon_b(k) + \frac{V}{2}n + \frac{V}{2}m \end{pmatrix}$$

cond. shift.

bare bands: $\epsilon_b(k) = \pm \frac{G}{2} - 2t_a(\cos(k_x) + \cos(k_y))$

$G \neq$ fixed gap! → see: <https://www.desmos.com/calculator/azrtesau5i>

bare band gap: $\epsilon_a(0) - \epsilon_b(0) = G - 2t_a(1+1) + 2t_b(1+1) = G - 4(t_a - t_b)$

↳ $t_a = -t_b \Rightarrow G_0 = G - 8t_a$

Hasse bands (= diry bras Δ) U_n v kodu

$\epsilon_a^H - \frac{V}{2} = \epsilon_a(k) \pm \frac{V}{2}m$

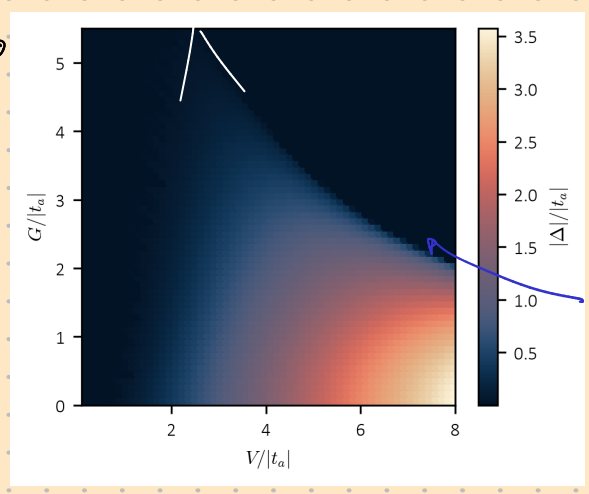
Hasse gap: $\epsilon_a^H(0) - \epsilon_b^H(0) = G - 8t_a - V \cdot m \equiv G_n$

note: $t_a \neq t_b$ is gap odpr pri $E_c = \frac{V}{2}n - \frac{t_a + t_b}{t_a - t_b} \cdot \frac{Vm - G}{2}$
 $V = U_n$
 odvisno od m

19. Za model z $U_n = V$ $\begin{pmatrix} \epsilon_a + U_n n_b & \dots \\ \dots & \epsilon_b + U_n n_a \end{pmatrix} \rightarrow \Delta = 0$ vedno za $\nexists$ EC režim!

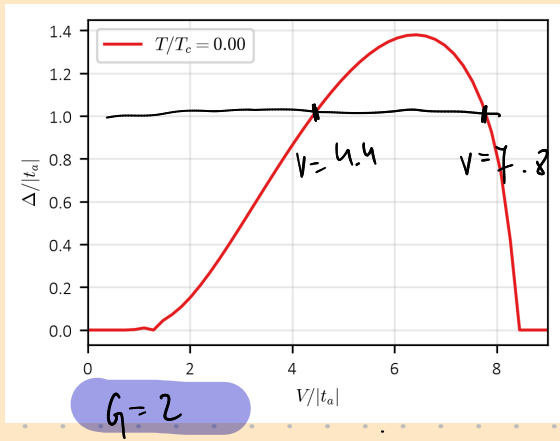
↳ če $U_n < V$ ($U_n = 0 =$ BCS teorija) $\Rightarrow \Delta > 0$ tudi za $G_0 > 0$.

$G_{max} \sim 5$
 za $U_n = V$



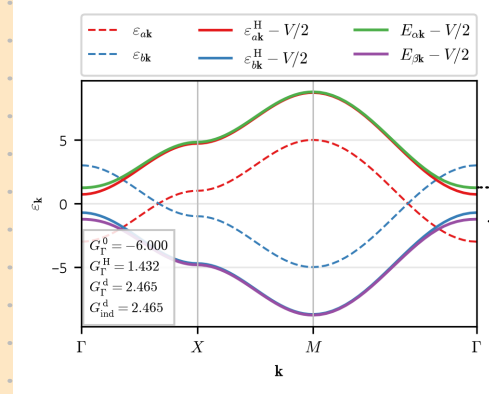
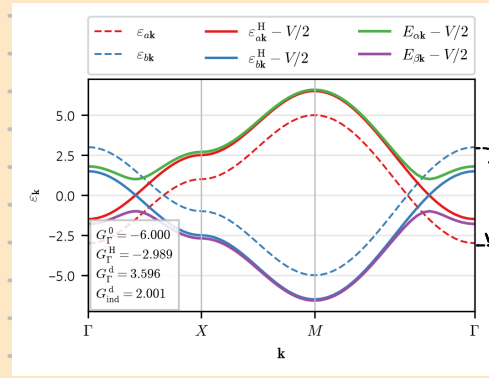
Hasse shift
 reko promoci
 za Δ , da
 ne razrije.
 $G_{max} \sim 5$
 Manjša z večim V , kar
 Hasse potegne navzgor.

prime 1: $V_H = V$:



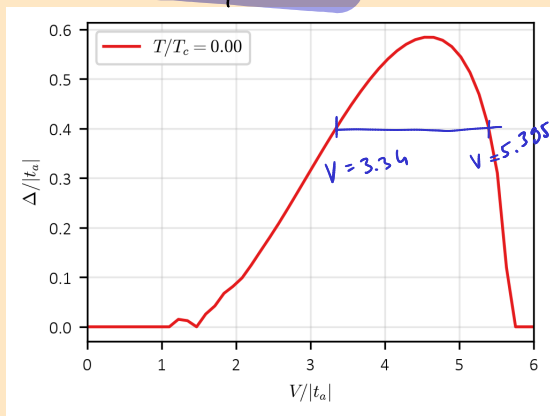
$V=4.4$

$V=7.8$



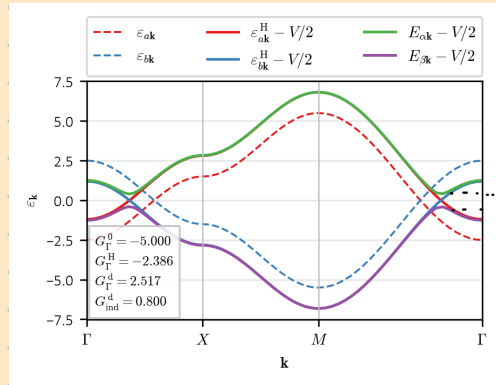
BEC-like (Hawke!)

$G=3$

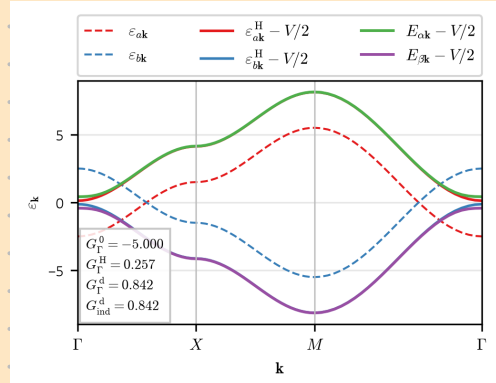


$V=3.34$

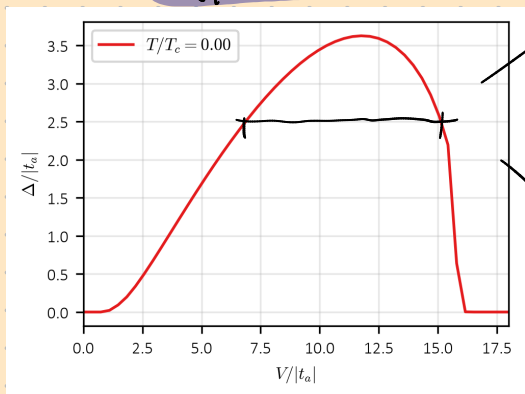
$V=5.4$



BUS

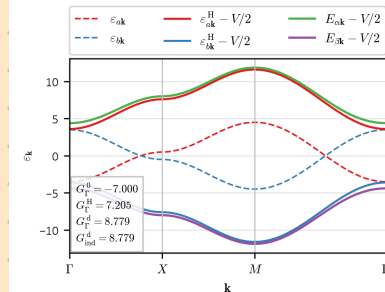
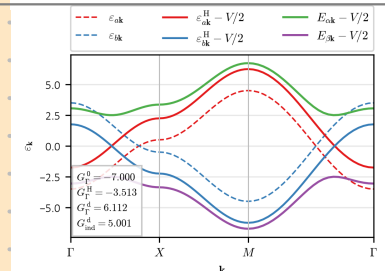


$G=10$



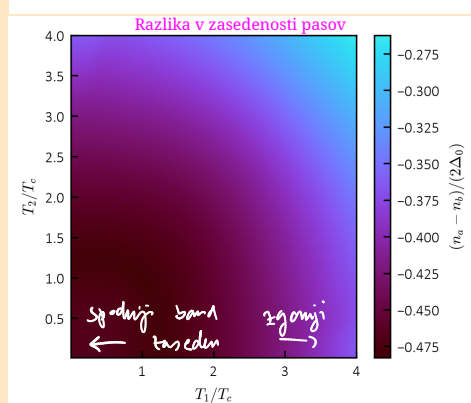
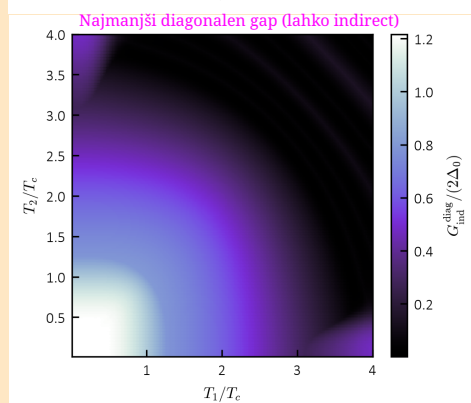
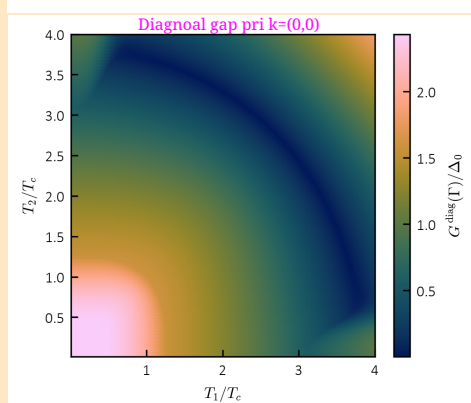
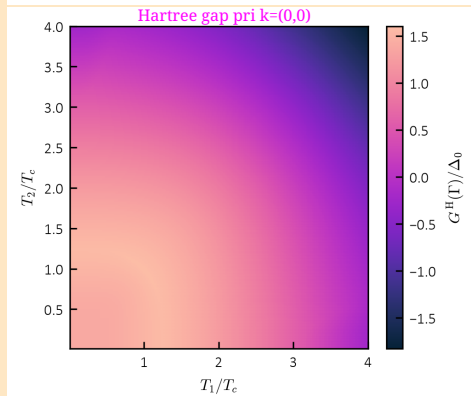
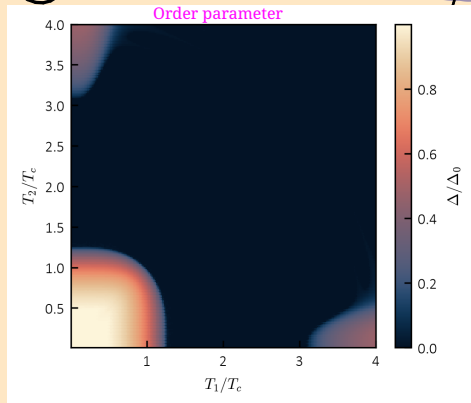
$V=6.866$

$V=15.13$

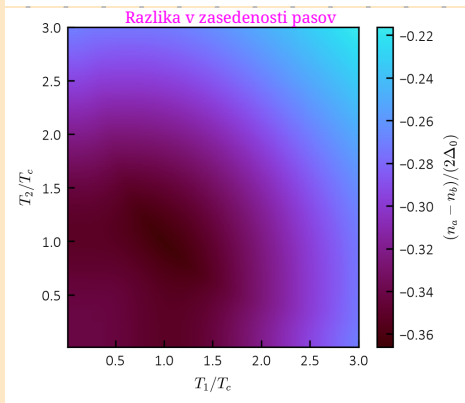
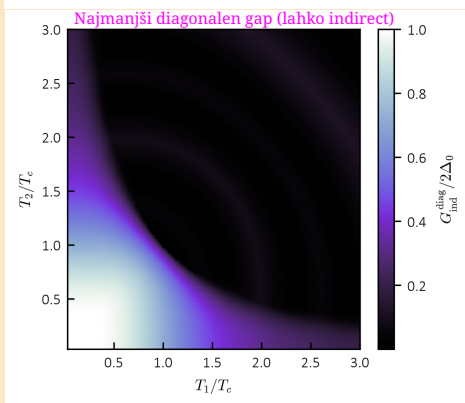
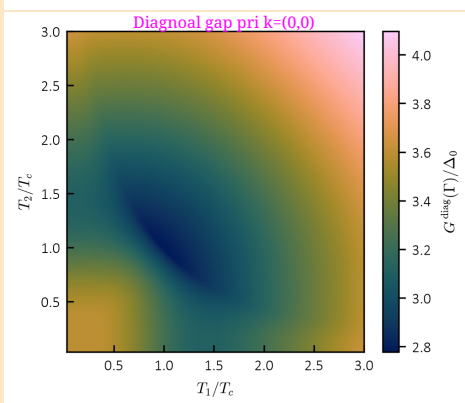
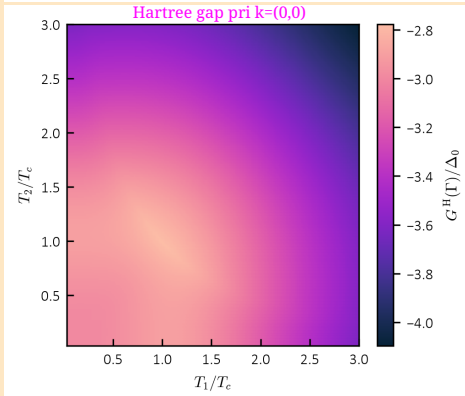
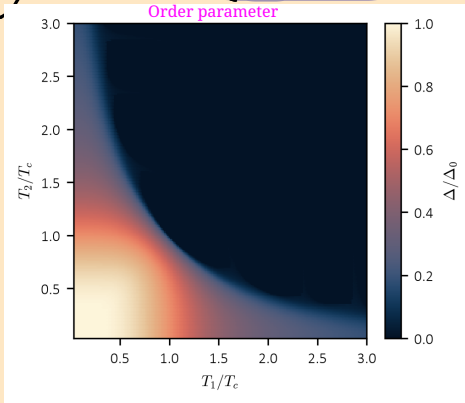


2. GAPS maps za : $G=2$, kopel: $\gamma(T) = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \cdot \text{r.i.} \cdot f\left(\frac{E_{\text{av}} - \frac{\gamma}{2}}{T}\right) f\left(-\frac{E_{\text{av}} - \frac{\gamma}{2}}{T}\right)$

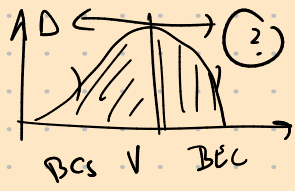
(A) $V=7.8$ (BEC r im):



(B) $V=4.4$ (BCS r im pri istem Δ)



3.) kaj se dogaja vmes?



okoli $\Delta \approx 0$:

$$\Delta' = \Delta \cdot \Lambda$$

$$\Lambda = V \sum_k w_k \mu_k v_k (n_{p_k} - n_{d_k}) = \frac{V}{N} \sum_k \frac{n_{p_k} - n_{d_k}}{E_{d_k} - E_{p_k}} (= |\sum_k| \cdot 2)$$

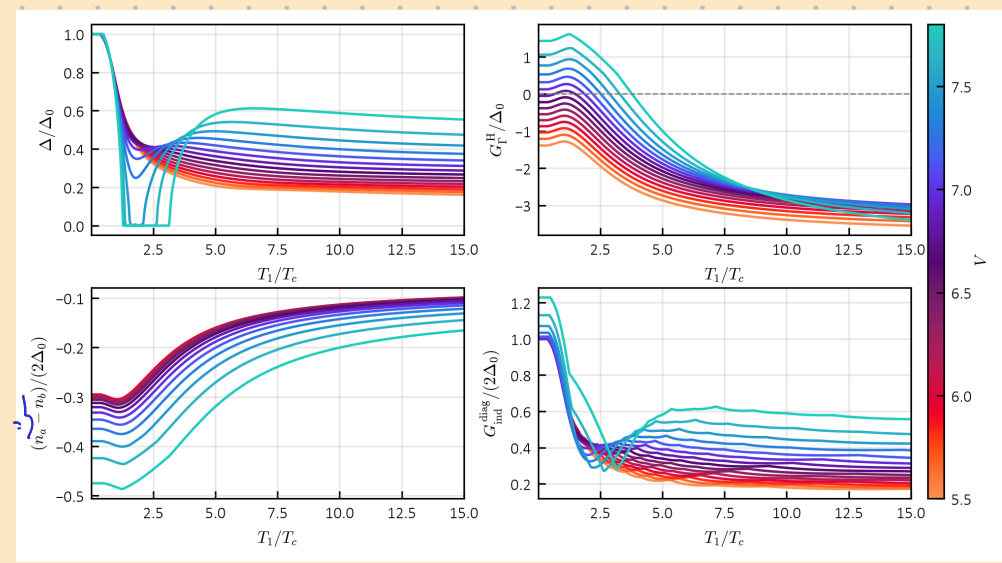
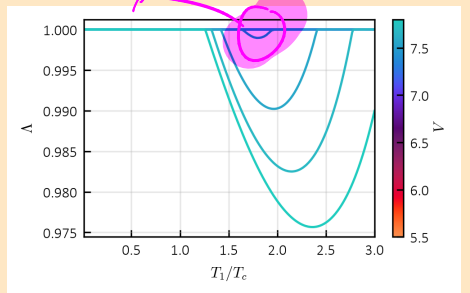
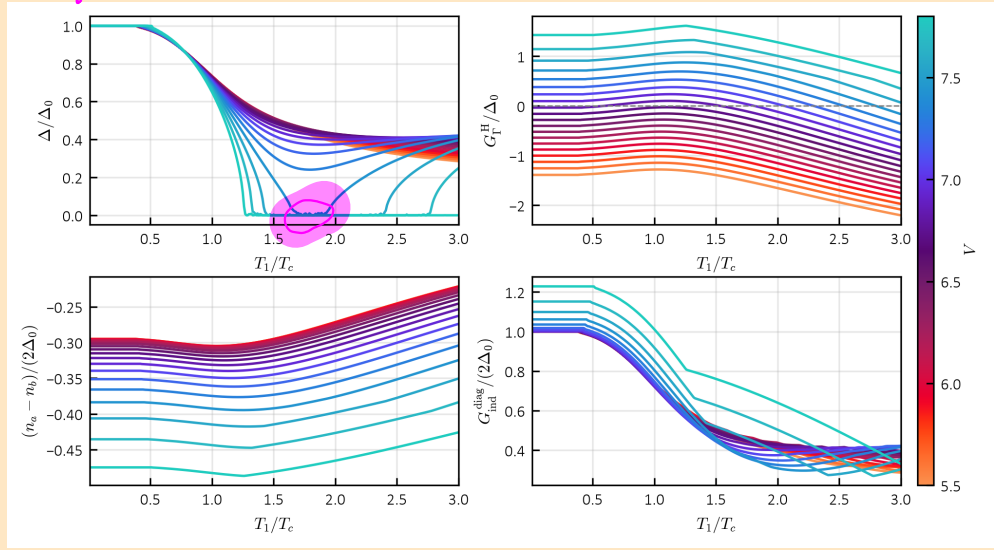
→ ko $\Lambda > 1$: normal state je unstable

→ $\Lambda = 1$ transition / kritična / fixed point

→ $\Lambda < 1$: $\Delta \rightarrow 0$, normal state.

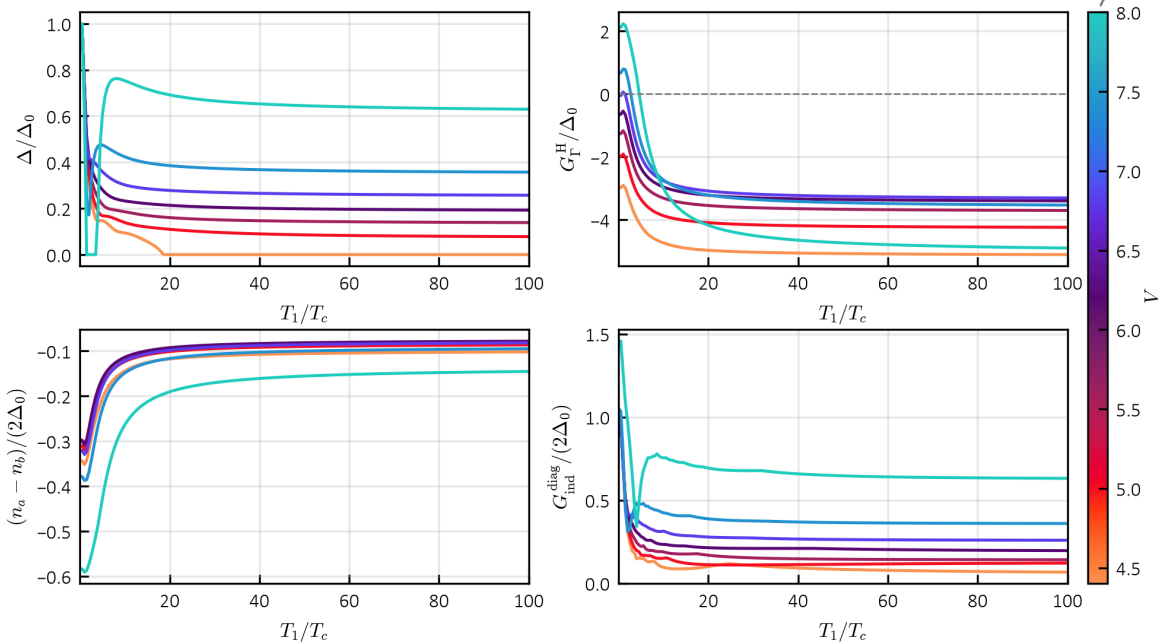
→ $|\Delta' - \Delta| = |\Lambda - 1| |\Delta|$ = gap residual, glej, da je za $|\Lambda - 1| \sim 0.1$ $|\Delta|$ nastavljen na $\sim 10^{-5}$.

$T_2 = 0.1 T_c$ fixed.

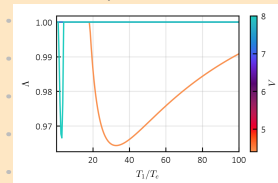


zakaj? $G^H = G^0 - Vm$ ko drizjemo T_1 : $n_p - n_d \rightarrow$ se manjša $\rightarrow \Lambda < 1 \rightarrow \Delta \rightarrow 0$
 ↓
 kerati: $m \uparrow \rightarrow G^H \downarrow \rightarrow$ manjši band separation $\rightarrow \Delta > 0?$
 $G^0 = -6$ na ryongjih grafih.

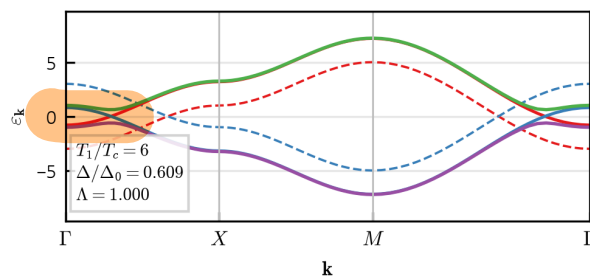
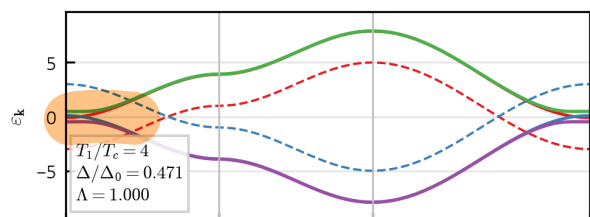
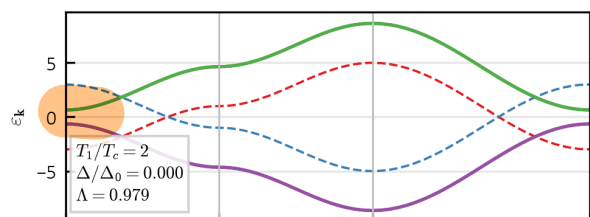
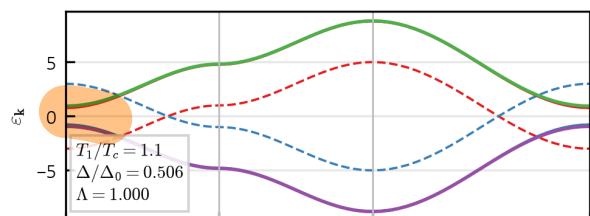
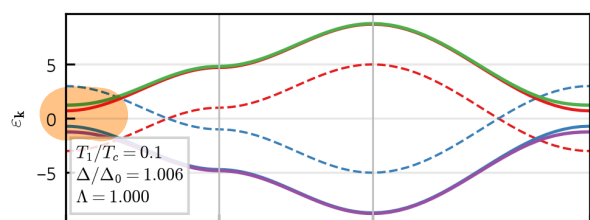
long sweep, D izgleda ravnino do $T_1 \rightarrow \infty$



v BCs režim
dijametro padne
0, ker $\Lambda < 1$



ϵ_{ak} (red dashed), ϵ_{bk} (blue dashed), $\epsilon_{ak}^H - V/2$ (red solid), $\epsilon_{bk}^H - V/2$ (blue solid), $E_{ak} - V/2$ (green solid), $E_{bk} - V/2$ (purple solid)



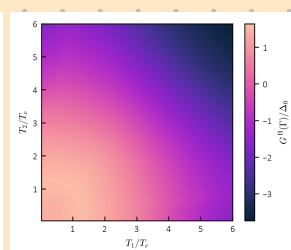
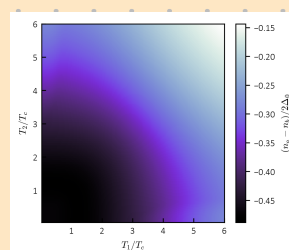
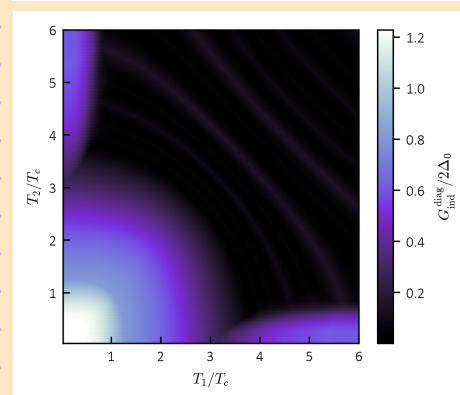
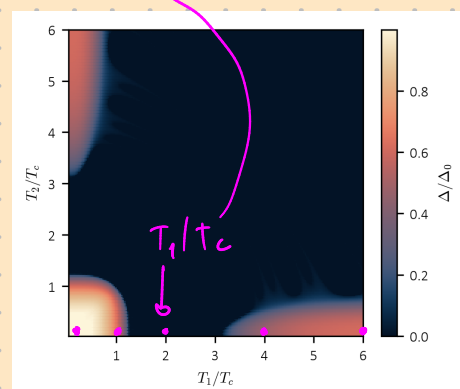
disperzija along $\Gamma-X-M$, za

$G=2$, $V=7.8$ pri $T_2=0.01T_c$
in različnih T_1 .

v pram otoku

$\Delta = 0$

reemergent
D, ker se
banda parietna
skupaj
(Nashre)



(4) equivalent T: (1) $\beta(0) \rightarrow n_{dk}(t) = \frac{1}{1 + \exp(\beta(t) E_{dk})} = f(E_{dk} \beta)$
 $(E - \frac{U}{2})?$

(2) $\langle \dot{H} \rangle = \sum_{k,j} E_{dk} \dot{n}_{dk} = \sum_{k,j} E_{dk} \sum_p [\Gamma_{kp}^{w)} n_{dp} (1 - n_{dk}) - \Gamma_{pk}^{dw} n_{dk} (1 - n_{dp})]$

(3) $\dot{\beta} = \frac{\langle \dot{H} \rangle_T}{\langle H^2 \rangle_T - \langle H \rangle_T^2}$

(4) $\beta \rightarrow \beta + \Delta t \dot{\beta}$

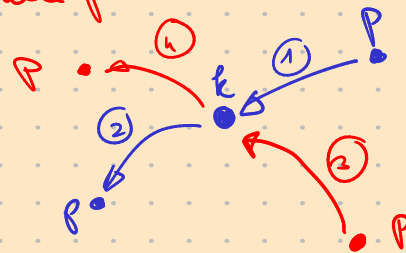
Interpretation: gley: /home/kzeleznikar/IJS-F1/Notes/theory/notes_open_systems_EI.pdf za podrobnejši.

$\downarrow \omega = \alpha$ fixed: *evoli?*

(*) $\dot{n}_{dk} = \sum_p \left[\begin{aligned} &\Gamma_{kp}^{\alpha\alpha} n_{dp} (1 - n_{dk}) \quad \textcircled{1} \\ &- \Gamma_{pk}^{\alpha\alpha} n_{dk} (1 - n_{dp}) \quad \textcircled{2} \\ &+ \Gamma_{kp}^{\alpha\beta} n_{\beta p} (1 - n_{dk}) \quad \textcircled{3} \\ &- \Gamma_{pk}^{\beta\alpha} n_{dk} (1 - n_{\beta p}) \quad \textcircled{4} \end{aligned} \right]$

delec α
 delec β

(p, k sistemano)

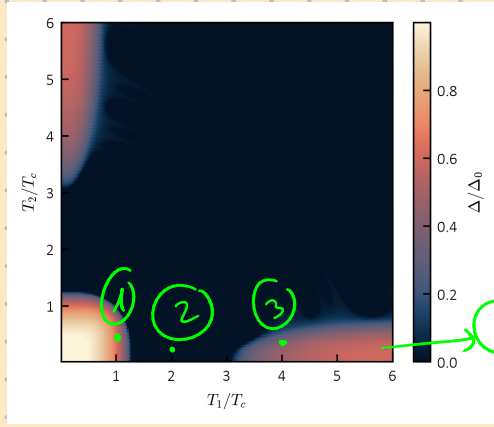


① $\propto$ iz $p \rightarrow k$ (ic ni zmedu)

② $\propto$ iz $k \rightarrow p$ (ic h j)

→ analiza fazdenesi delov za $q=2$

(A) BEC režim ($v=7.8$)



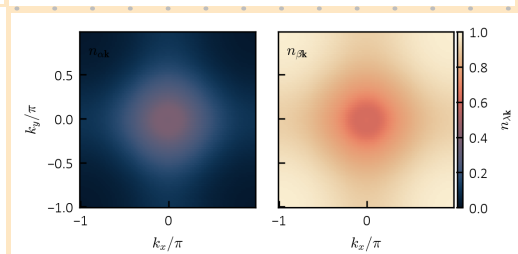
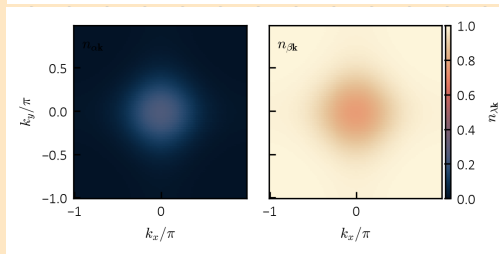
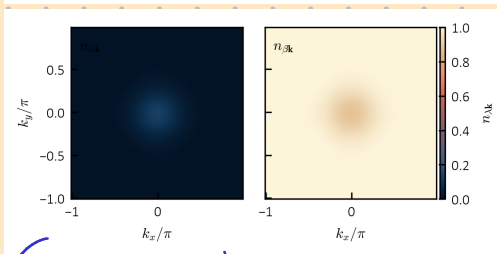
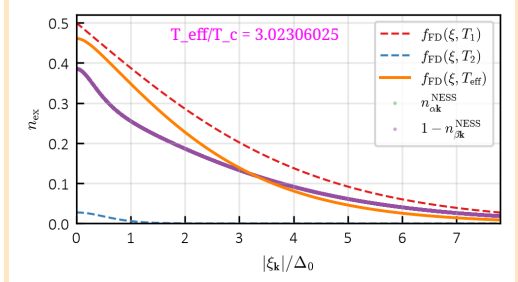
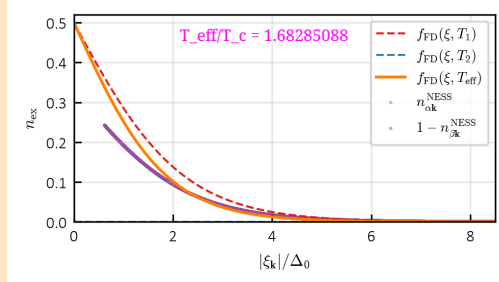
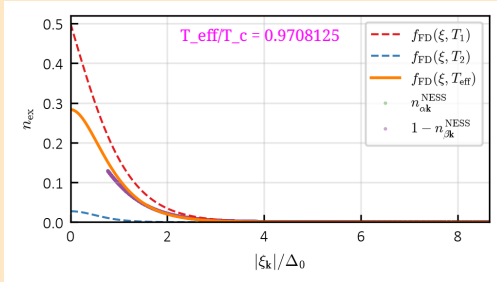
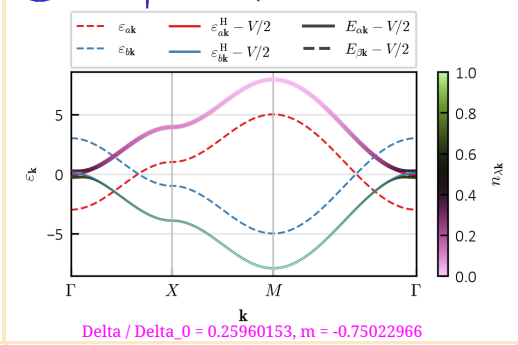
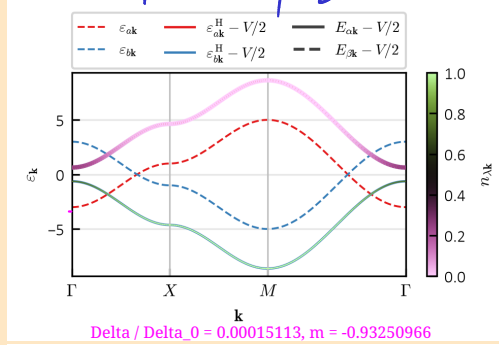
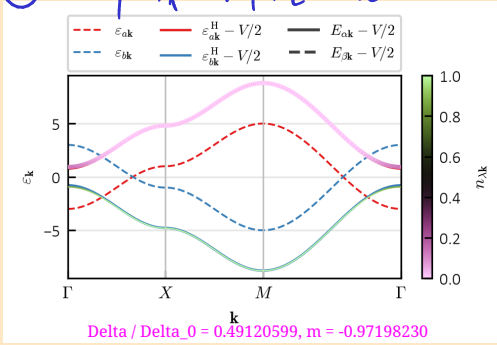
$$(\text{opomba } \xi_k = \frac{E_a(k) - E_v(k) - v(n_a - n_b)}{2})$$

$$E_{ak} - \frac{V}{2} = \frac{E_{ak} + \xi_k}{2} \pm \sqrt{\xi_k^2 + D^2}$$

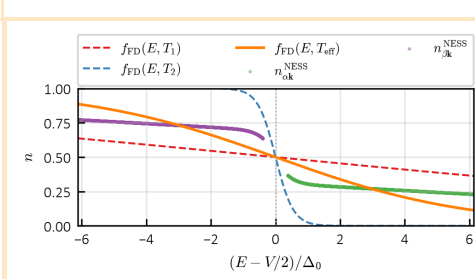
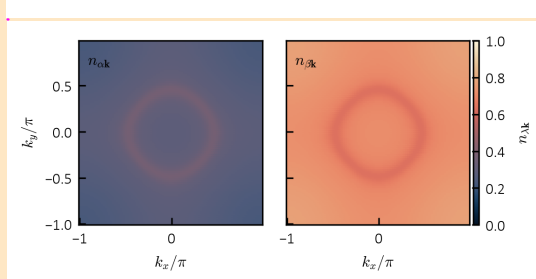
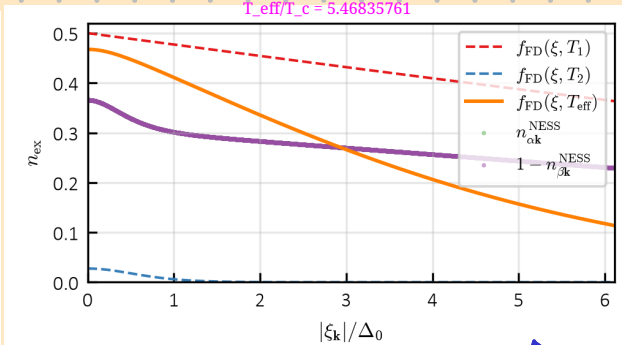
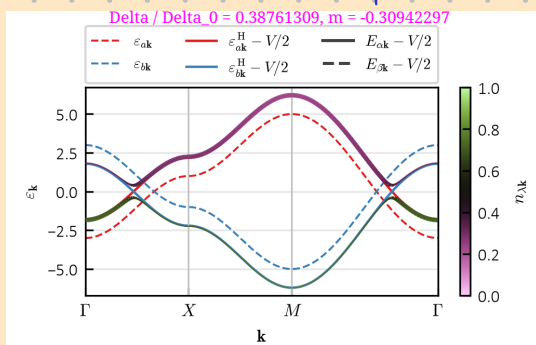
① BEC / $T_1 = 1.1 / T_2 = 0.5$

② $D=0$ / $T_1 = 2 / T_2 = 0.2$

③ BCS / $T_1 = 4 / T_2 = 0.5$



(4) BCS / $T_1 = 20 / T_2 = 0.5$

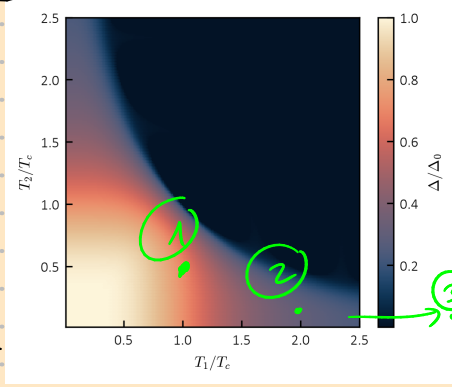


isto, drugi
variable!

→ ko $T_1 \rightarrow \infty, T_c \rightarrow 0$ gre

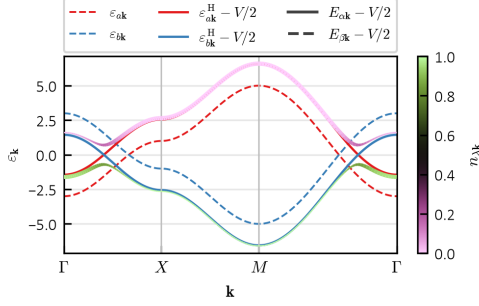
$n_k \rightarrow$

(B) BCS limit ($V=4.4$, $g=2$)



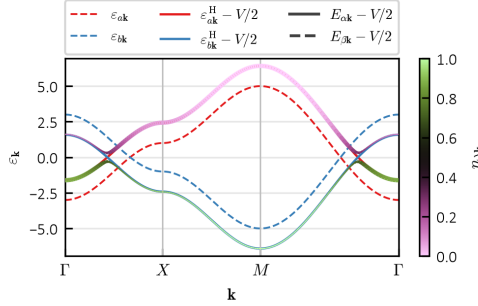
1) $T_1 = 1.1$ / $T_2 = 0.5$

Delta / Delta_0 = 0.62227223, m = -0.70813060



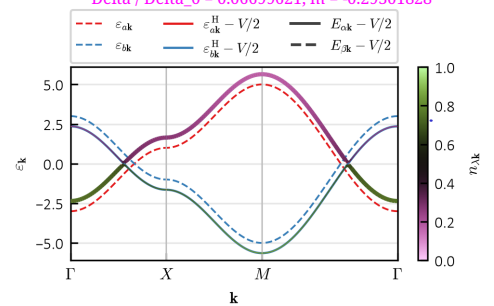
2) $T_1 = 2$ / $T_2 = 0.2$

Delta / Delta_0 = 0.29625834, m = -0.64265174

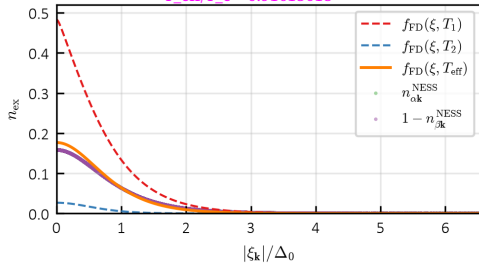


3) $T_1 = 10$, $T_2 = 0.1$

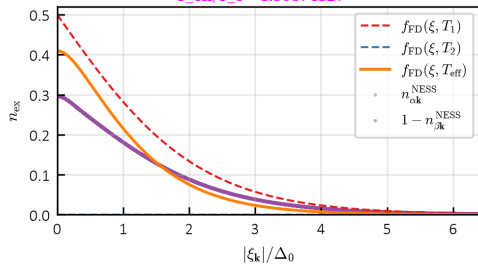
Delta / Delta_0 = 0.06699621, m = -0.29301828



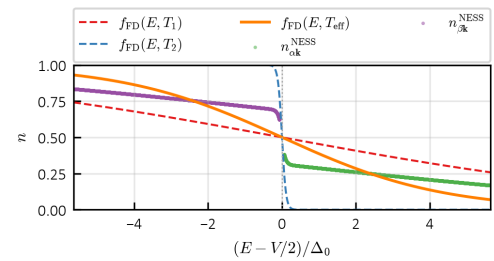
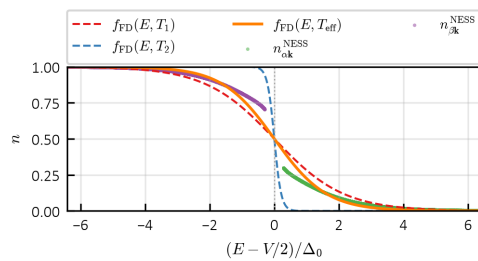
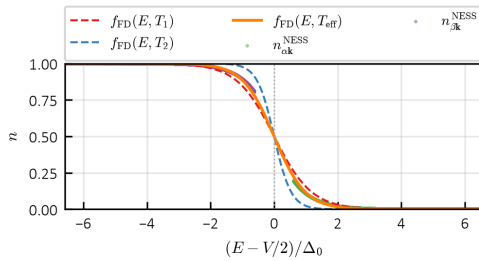
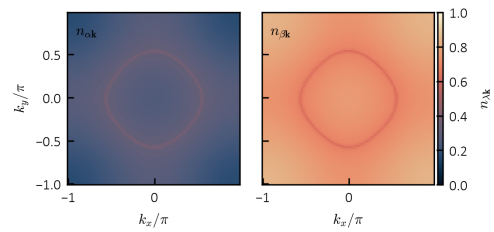
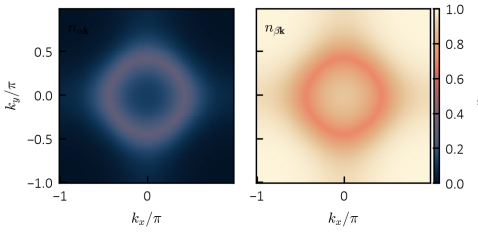
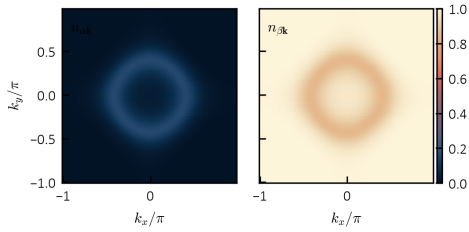
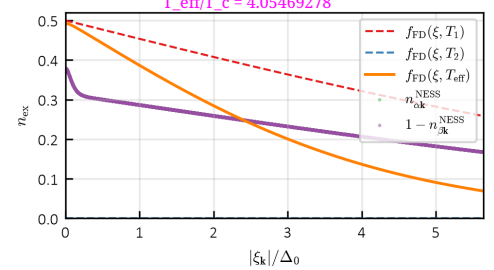
T_eff / T_c = 0.91613618



T_eff / T_c = 1.50874127



T_eff / T_c = 4.05469278



Technical notes:

Kopel za vsa ploste uporablja oblike: $f_{kp}^{(\mu\nu)} = 0.1 \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}^{\alpha} \cdot f\left(\frac{E_{\mu\nu} - \frac{V}{2}}{T}\right) f\left(-\frac{E_{\nu\mu} - \frac{V}{2}}{T}\right)$

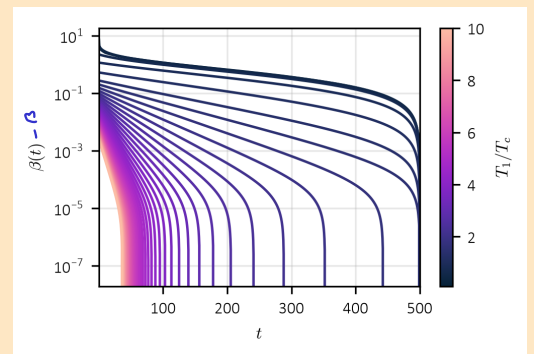
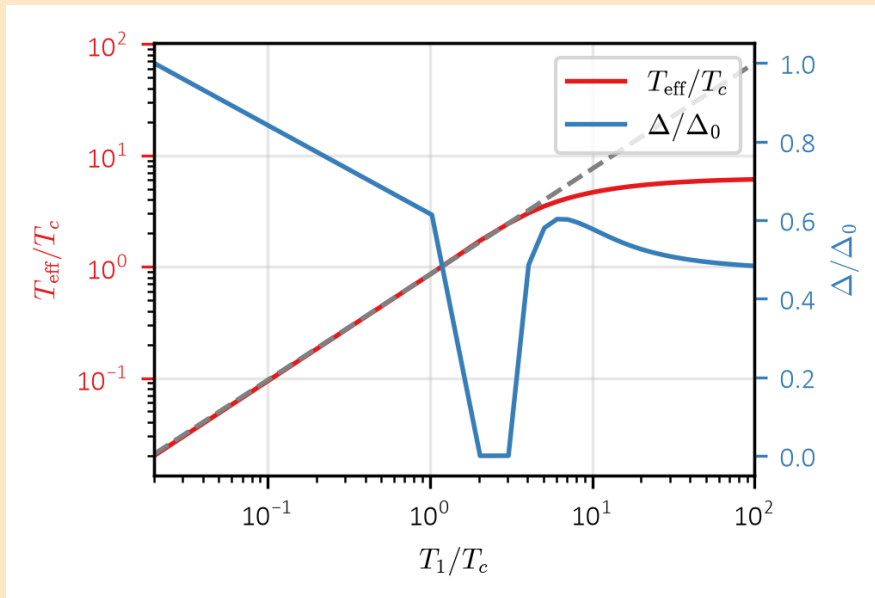
$$\bar{L}_{kp}^{(\mu\nu)} = f_{kp}^{(\mu\nu)} d_{\mu k}^{\dagger} d_{\nu p} \quad (\text{number conserving bath})$$

→ tok tokrega L je enak (*)

oznake: $H^{MF} = E_0 + \sum_k (c_{\alpha k}^{\dagger}, c_{\beta k}^{\dagger}) \begin{pmatrix} E_{\alpha k} + \frac{V}{2} n - \frac{V}{2} m & -\Delta^* \\ -\Delta & E_{\beta k} + \frac{V}{2} n + \frac{V}{2} m \end{pmatrix} \begin{pmatrix} c_{\alpha k} \\ c_{\beta k} \end{pmatrix}$

$$\hookrightarrow E_{\alpha k} = \gamma_k \pm \sqrt{\xi_k^2 + \Delta^2}; \quad \gamma_k = \frac{E_{\alpha} + E_{\beta} + V \cdot n}{2} \quad \xi_k = \frac{E_{\alpha} + E_{\beta} - V \cdot m}{2}$$

$$E_{\alpha, \beta}^H = E_{\alpha, \beta} + \frac{V}{2} n \mp \frac{V}{2} m$$



Crossing energies in $t_a \neq -t_b$: (Hartree)

$$E_a^H = E_b^H \Rightarrow \frac{G}{2} - 2t_a(c_x + c_y) + \frac{Vn}{2} - \frac{Vm}{2} = -\frac{G}{2} - 2t_b(c_x + c_y) + \frac{Vn}{2} + \frac{Vm}{2}$$

$$\Rightarrow c_x + c_y = \frac{G - Vm}{2(t_a - t_b)}$$

$$\mu_B = \frac{G}{2} - \frac{t_a}{t_a - t_b} (G - Vm) - \frac{V}{2} m + \frac{Vn}{2} = \frac{Vn}{2} + (G - Vm) \left(\frac{1}{2} - \frac{t_a}{t_a - t_b} \right)$$

$\frac{t_a - t_b - 2t_a}{2(t_a - t_b)}$

in $t_a = -t_b$ ✓

$$\Rightarrow \mu_B = \frac{Vn}{2} - \frac{t_a + t_b}{t_a - t_b} \cdot \frac{G - Vm}{2}$$

→ Δ_0 : mit: $D(t=0) = 1$, $m(t=0) = 0$, $\mu(t=0) = 0$

$\Rightarrow$ wichtige: $n = 1 = \sum_k w_k (f(\epsilon_{\alpha k} - \mu) + f(\epsilon_{\beta k} - \mu)) \Rightarrow \mu$
 $\Rightarrow n_{\alpha k} = \frac{n_k^2}{n^2} n_{\alpha k} + \frac{n_k^2}{n^2} n_{\beta k}$
 $\Rightarrow D' = V \sum_k w_k n_k v_k (n_{\beta k} - n_{\alpha k})$
 $\Rightarrow m' = \sum_k w_k (n_{\alpha k} - n_{\beta k})$
 $\Rightarrow \Delta \leftarrow \Delta + 0.15 (D' - \Delta)$
 $m \leftarrow m + 0.15 (m' - m)$

while $\hat{r} = \max(D - D', m - m', n - n')$ > tol
 max

→ T_c : fix $D = 0 \rightarrow$ solve for m, μ :

$$D = V \sum_k w_k \frac{D}{2|\epsilon_k|} (n_{\beta k} - n_{\alpha k})$$

$$\Rightarrow \mu = \sum_k w_k \frac{(n_{\beta k} - n_{\alpha k})}{\epsilon_{\alpha k} - \epsilon_{\beta k}} \quad (T)$$

→ $\Delta > 1 \Leftrightarrow D > 0$

→ $\Delta = 1 \Leftrightarrow T_c$

→ $\Delta < 1 \Leftrightarrow D < 0$

→ set $T_{\min}, T_{\max} \rightarrow$ restrict Δ .

→ open system: Ver. ist ket closed, same: $n_{\alpha\beta} \neq f \rightarrow$ Se eine matrix zu $n_{\alpha/\beta}$:

$\rightarrow$ zu $n_{\alpha\beta}$ in themal jump
 $\Rightarrow E_{\alpha\beta}, m_{\alpha}, n_{\alpha}$
 $\Rightarrow \dot{n}_i = (1 - n_i) \sum_j \Gamma_{ij} n_j w_j - n_i \sum_j \Gamma_{ji} (1 - n_j)$
 $\Rightarrow n_{\alpha k} \leftarrow n_{\alpha k} + h \cdot \dot{n}_{\alpha k} \Rightarrow \sum_k w_k (n_{\alpha k} + n_{\beta k}) = n \Rightarrow f_i$ kann $n_{\beta}(t+h)$
 $\Rightarrow$ stable value ket thermal: zeichne Δ, m

while $\hat{r} = \max(D - D', m - m', n - n', \max_{i,k} (n'_{i,k})) > \text{tol}$